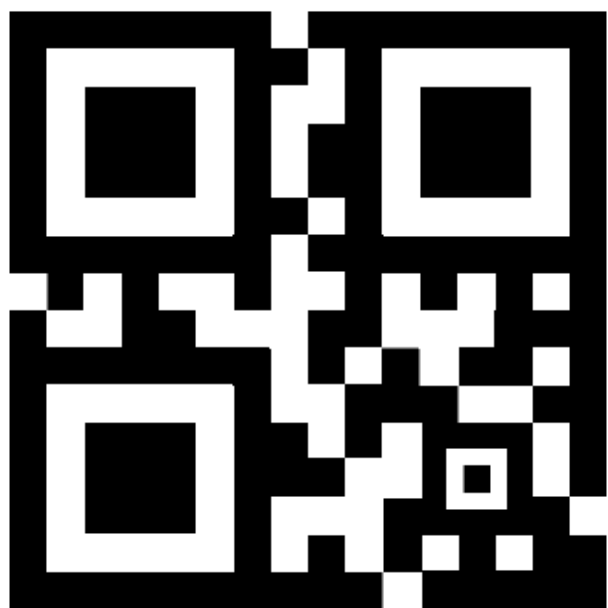




LabTrackQR

User Manual

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Welcome to the LabTrackQR tracking system.

This guide will walk you through everything you need to know to use the client application efficiently, from basic login to advanced map tracking and sample history management.

Chapter 1: Application Basics & Interface

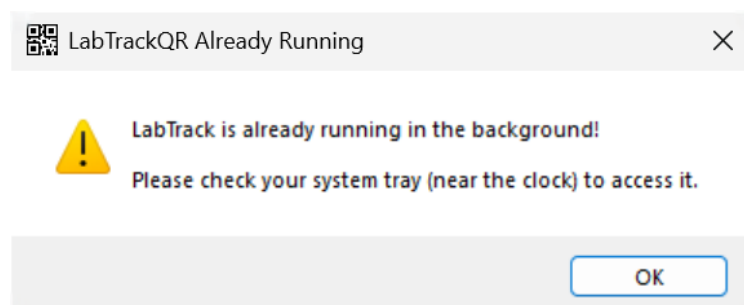
LabTrackQR is designed to be unobtrusive. It runs silently in the background of your computer, waking up instantly only when you scan a barcode or request to view data.

1.1 Starting the App

When you launch LabTrackQR, you will briefly see the startup sequence as the application connects to the hardware scanner and the master server.



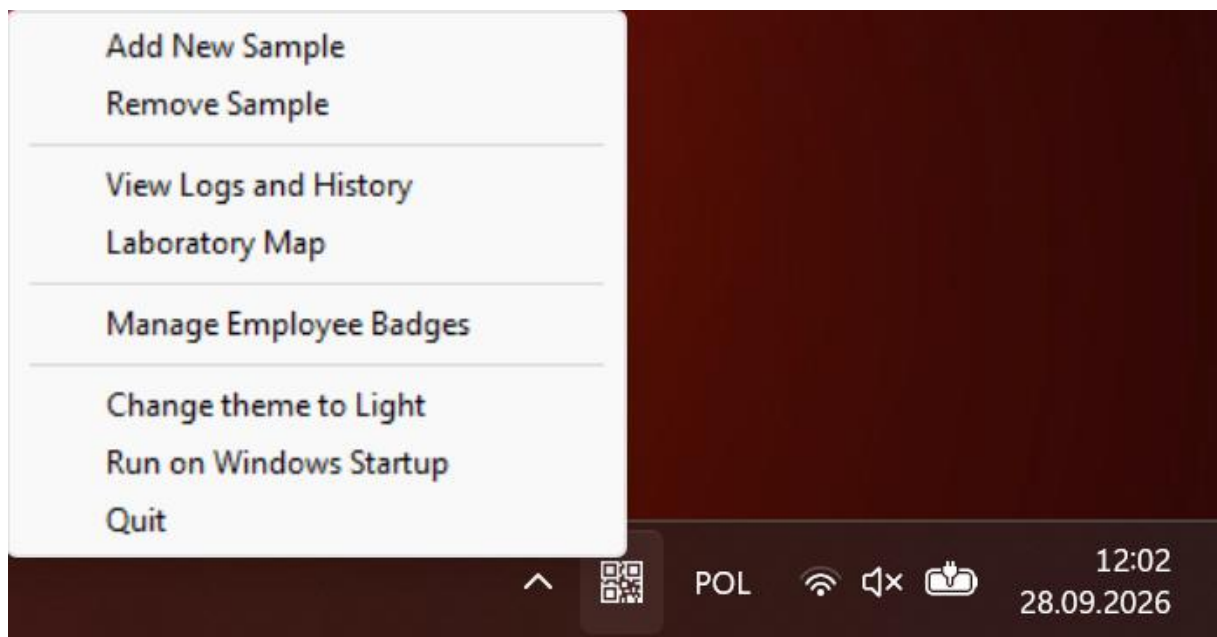
Because LabTrackQR runs in the background, clicking the desktop icon while the app is already open will not open a new window. Instead, the system will prevent duplicate instances and remind you to check your system tray.



1.2 The System Tray & Menu

Once running, LabTrackQR lives in your Windows System Tray (the area next to your clock in the bottom right corner of the screen).

Look for the LabTrackQR icon. To access the main controls, simply Right-Click this icon.

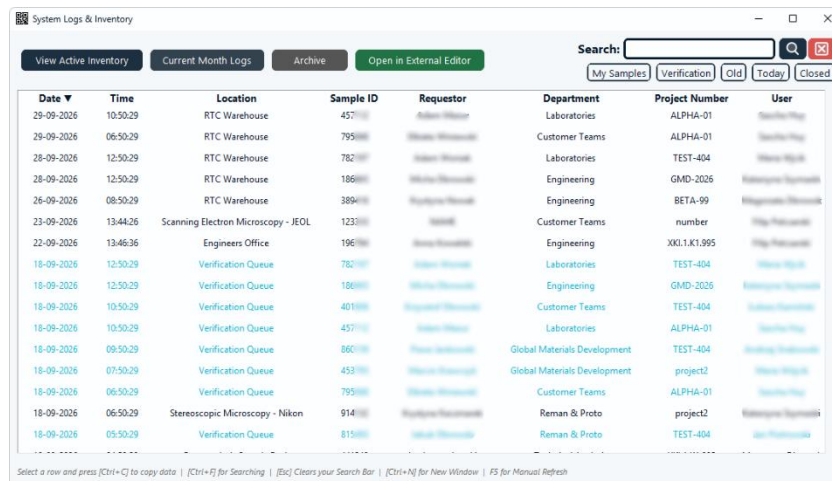


This tray menu is your main control hub. From here, you can open all major viewers, initialize manual samples, or adjust application settings like enabling it to start automatically with Windows.

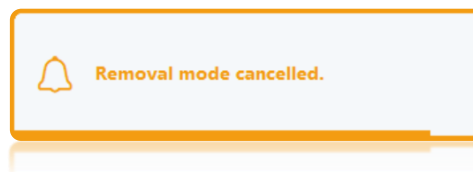
1.3 Theme Options

LabTrackQR fully supports both Light and Dark modes. By default, it will try to match your Windows system theme, but you can manually force your preferred visual style directly from the Tray Menu.

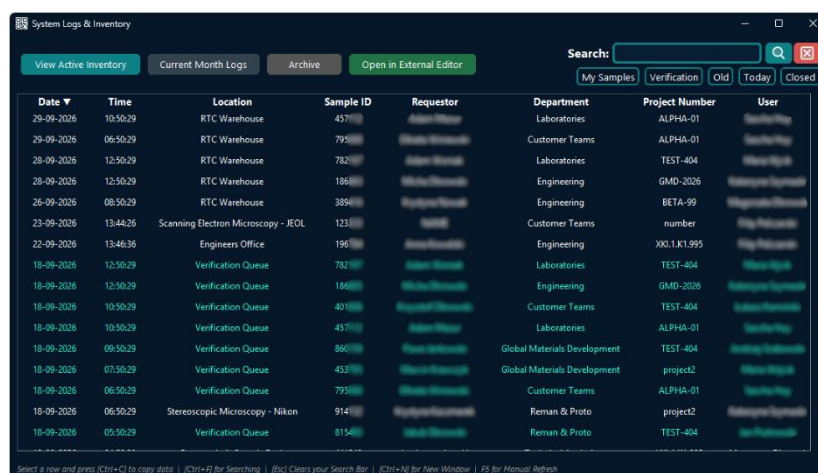
Light theme:



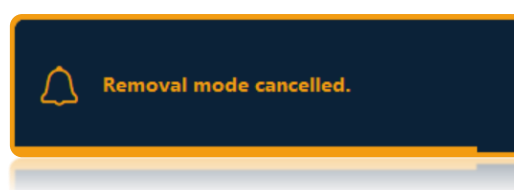
Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
29-09-2026	10:50:29	RTC Warehouse	457	Adam Wilson	Laboratories	ALPHA-01	David Ray
29-09-2026	06:50:29	RTC Warehouse	795	Adam Wilson	Customer Teams	ALPHA-01	David Ray
28-09-2026	12:50:29	RTC Warehouse	782	Adam Wilson	Laboratories	TEST-404	Adam Wilson
28-09-2026	12:50:29	RTC Warehouse	106	Adam Wilson	Engineering	GMD-2026	Adam Wilson
26-09-2026	08:50:29	RTC Warehouse	389	Adam Wilson	Engineering	BETA-99	Adam Wilson
23-09-2026	13:44:26	Scanning Electron Microscopy - JEOL	123	Adam Wilson	Customer Teams	number	Adam Wilson
22-09-2026	13:46:36	Engineers Office	196	Adam Wilson	Engineering	XX01.K1.995	Adam Wilson
18-09-2026	12:50:29	Verification Queue	782	Adam Wilson	Laboratories	TEST-404	Adam Wilson
18-09-2026	12:50:29	Verification Queue	106	Adam Wilson	Engineering	GMD-2026	Adam Wilson
18-09-2026	10:50:29	Verification Queue	401	Adam Wilson	Customer Teams	TEST-404	Adam Wilson
18-09-2026	10:50:29	Verification Queue	457	Adam Wilson	Laboratories	ALPHA-01	David Ray
18-09-2026	09:50:29	Verification Queue	860	Adam Wilson	Global Materials Development	TEST-404	Adam Wilson
18-09-2026	07:50:29	Verification Queue	453	Adam Wilson	Global Materials Development	project2	Adam Wilson
18-09-2026	06:50:29	Verification Queue	795	Adam Wilson	Customer Teams	ALPHA-01	David Ray
18-09-2026	06:50:29	Stereoscopic Microscopy - Nikon	914	Adam Wilson	Reman & Proto	project2	Adam Wilson
18-09-2026	05:50:29	Verification Queue	815	Adam Wilson	Reman & Proto	TEST-404	Adam Wilson



Dark theme:



Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
29-09-2026	10:50:29	RTC Warehouse	457	Adam Wilson	Laboratories	ALPHA-01	David Ray
29-09-2026	06:50:29	RTC Warehouse	795	Adam Wilson	Customer Teams	ALPHA-01	David Ray
28-09-2026	12:50:29	RTC Warehouse	782	Adam Wilson	Laboratories	TEST-404	Adam Wilson
28-09-2026	12:50:29	RTC Warehouse	106	Adam Wilson	Engineering	GMD-2026	Adam Wilson
26-09-2026	08:50:29	RTC Warehouse	389	Adam Wilson	Engineering	BETA-99	Adam Wilson
23-09-2026	13:44:26	Scanning Electron Microscopy - JEOL	123	Adam Wilson	Customer Teams	number	Adam Wilson
22-09-2026	13:46:36	Engineers Office	196	Adam Wilson	Engineering	XX01.K1.995	Adam Wilson
18-09-2026	12:50:29	Verification Queue	782	Adam Wilson	Laboratories	TEST-404	Adam Wilson
18-09-2026	12:50:29	Verification Queue	106	Adam Wilson	Engineering	GMD-2026	Adam Wilson
18-09-2026	10:50:29	Verification Queue	401	Adam Wilson	Customer Teams	TEST-404	Adam Wilson
18-09-2026	10:50:29	Verification Queue	457	Adam Wilson	Laboratories	ALPHA-01	David Ray
18-09-2026	09:50:29	Verification Queue	860	Adam Wilson	Global Materials Development	TEST-404	Adam Wilson
18-09-2026	07:50:29	Verification Queue	453	Adam Wilson	Global Materials Development	project2	Adam Wilson
18-09-2026	06:50:29	Verification Queue	795	Adam Wilson	Customer Teams	ALPHA-01	David Ray
18-09-2026	06:50:29	Stereoscopic Microscopy - Nikon	914	Adam Wilson	Reman & Proto	project2	Adam Wilson
18-09-2026	05:50:29	Verification Queue	815	Adam Wilson	Reman & Proto	TEST-404	Adam Wilson



Chapter 2: User Login & Authentication

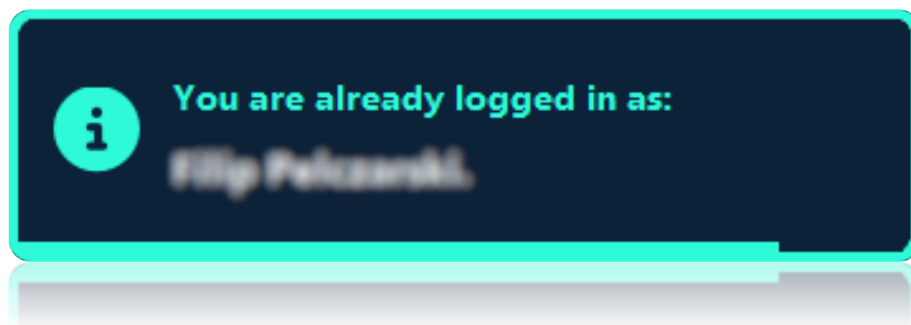
To ensure accountability, all actions in LabTrackQR are tied to the active user. You must be logged in to a scanner to perform actions.

2.1 Standard Login

To log in, simply scan your personal Employee QR Badge using the attached barcode scanner. If the login is successful, a notification will slide onto your screen confirming your active session.

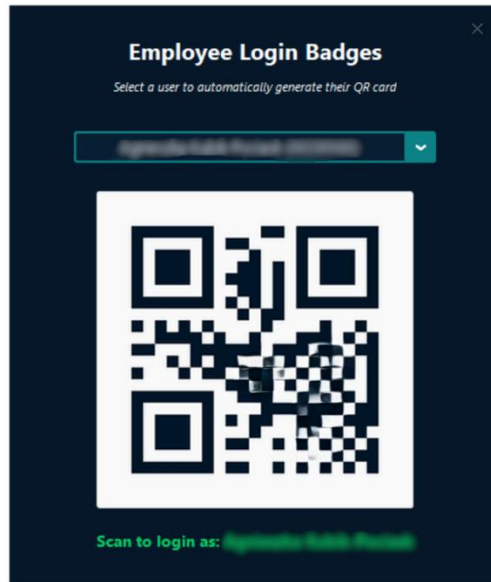


If you accidentally scan your badge again while already logged in, the system will simply remind you of your current active session.

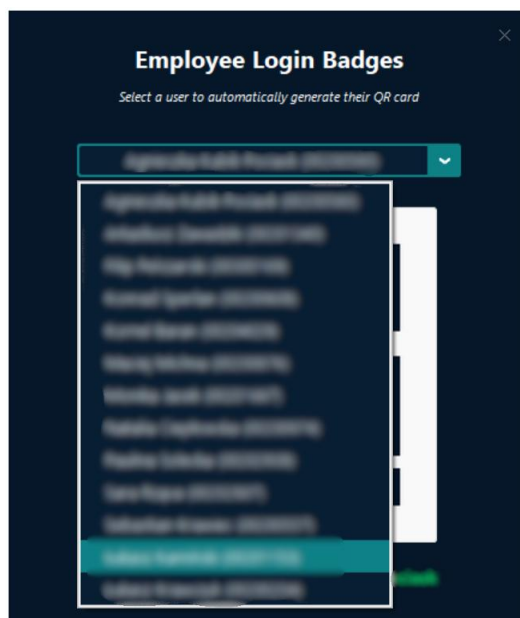


2.2 Generating Employee Badges

If you need a physical QR code to log in, you can generate one directly from the app. Open the Tray Menu and select Manage Employee Badges.



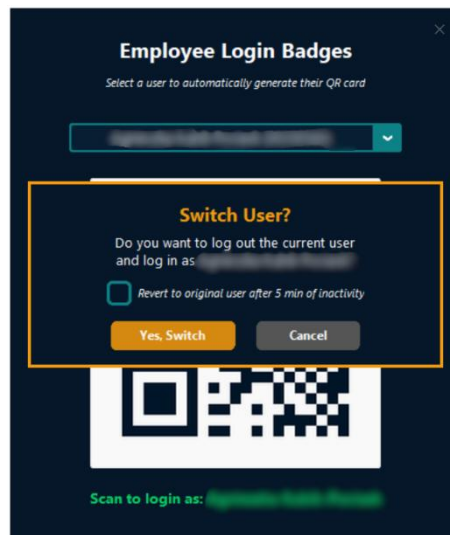
Use the dropdown menu to find your name in the registered employee database.



Once selected, the system will generate a secure QR code on your screen. You can scan this directly from the monitor to log in!

2.3 Switching Users (Relogging)

If someone else is currently logged into the scanner but you need to quickly process a sample under your own name, simply scan your badge. The system will detect the user change and prompt a Switch User confirmation.



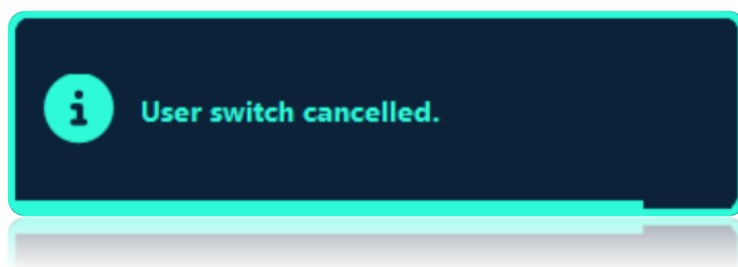
The 5-Minute Idle Timer (Temporary Login)

When switching users, you have the option to check a box that says "Revert to original user after 5 min of inactivity".

- If **checked**: You will be logged in temporarily. If you don't scan anything for 5 minutes, the system will automatically log you out and restore the previous user's session.
- If **unchecked**: You will permanently take over the scanner session.



If you scan your badge by mistake and do not want to take over the session, simply click Cancel on the popup, or wait 15 seconds for it to cancel automatically.



2.4 New AD User Registration

If you scan a brand-new badge that the system does not recognize, and you are logged into a Windows account, LabTrackQR will prompt you to register the badge.

[Insert Screenshot: new ad user registration popup]

Fill in your First and Last name. The system will automatically link this new badge to your current Windows Active Directory (AD) account, so in the future, the app can recognize you even without your badge!

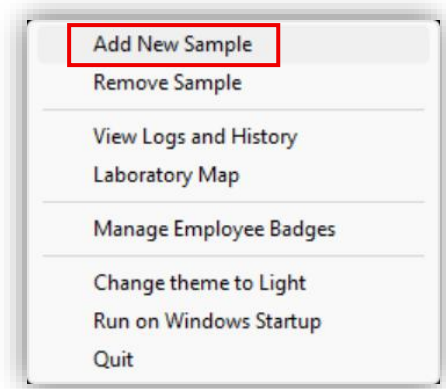
Chapter 3: Sample Tracking (The Core Workflow)

The primary function of LabTrackQR is pairing a physical sample with its laboratory location. The system is designed to be incredibly fast and flexible, adapting to how you prefer to scan.

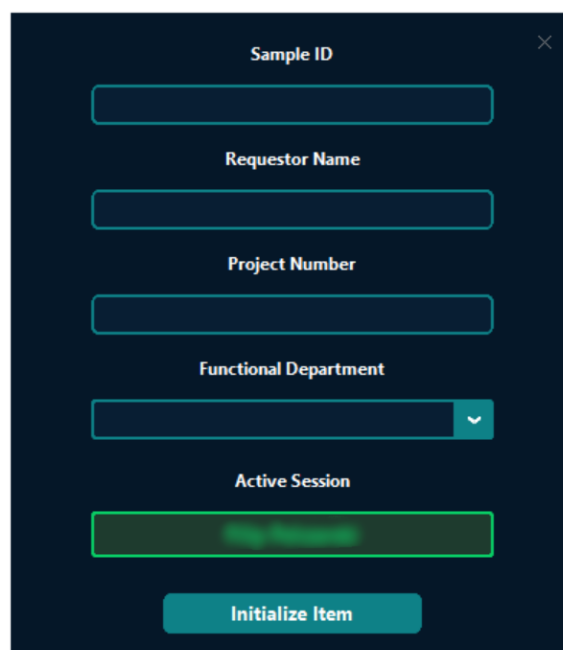
3.1 Manual Sample Entry

If a sample barcode is damaged, or you need to initialize a brand-new sample into the system without the scanner, you can enter it manually.

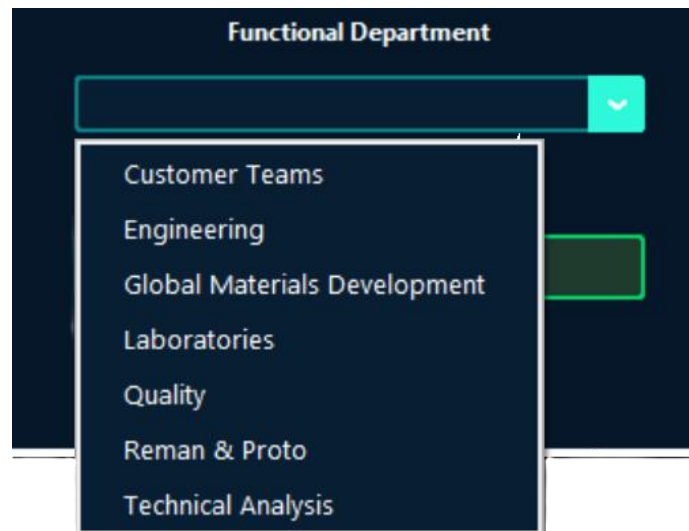
First, open the System Tray menu and select **Add New Sample**.



This will open the Manual Sample Entry form.

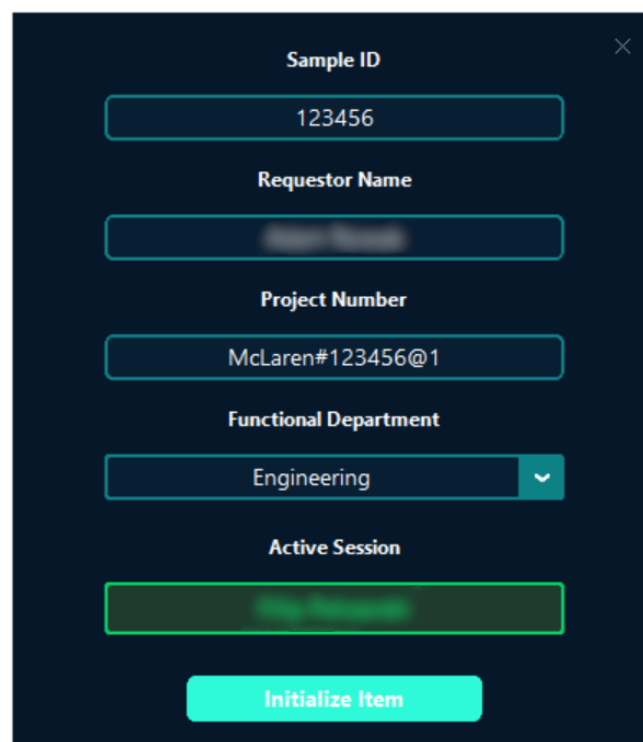
A screenshot of a web-based form titled 'Sample ID' with a close button (X) in the top right corner. The form has a dark blue background with teal-colored input fields and labels. The fields are: 'Requestor Name', 'Project Number', 'Functional Department' (a dropdown menu), and 'Active Session' (a text area with a green border). At the bottom of the form is a teal button labeled 'Initialize Item'.

Fill out all the required fields. You must provide the exact Sample ID, Requestor Name, and Project Number. Use the dropdown menu to select the proper Functional Department.



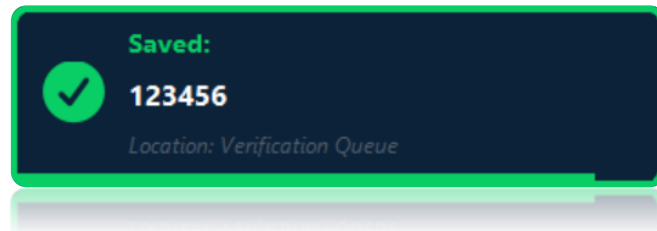
A screenshot of a dropdown menu titled "Functional Department". The menu is open, showing a list of options: Customer Teams, Engineering, Global Materials Development, Laboratories, Quality, Reman & Proto, and Technical Analysis. The dropdown is set against a dark blue background.

Make sure the "Active Session" at the bottom shows your name. If multiple users are logged in, use the dropdown to ensure the sample is credited to the correct person.



A screenshot of a form titled "Sample ID" with a close button (X) in the top right corner. The form contains the following fields: "Sample ID" with the value "123456", "Requestor Name" with a blurred value, "Project Number" with the value "McLaren#123456@1", "Functional Department" with a dropdown menu showing "Engineering", and "Active Session" with a blurred value. At the bottom of the form is a red button labeled "Initialize Item".

Once you click **Initialize Item**, the system will automatically place the sample into the default **Verification Queue** location. You will receive a success popup confirming it was saved.

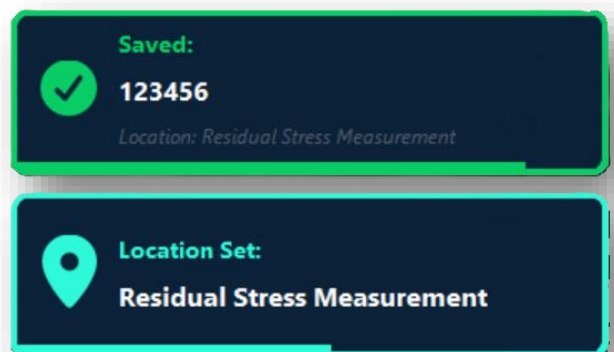


3.2 Standard Hardware Scanning

Using the physical barcode scanner is the fastest way to log data. The app supports two different scanning workflows:

Workflow A: Location First (Standard)

Scan a Location barcode (starting with LOC:). The system will set this as your active location. Then, scan a Sample barcode (starting with SMP:). The sample will instantly be saved to that location.



Workflow B: Sample First (Queuing)

If you scan a Sample (SMP:) before scanning a location, the app will safely place it into a "Pending Queue". You can queue multiple samples this way. Once you finally scan a Location (LOC:), the system will automatically assign all queued samples to that location at once!



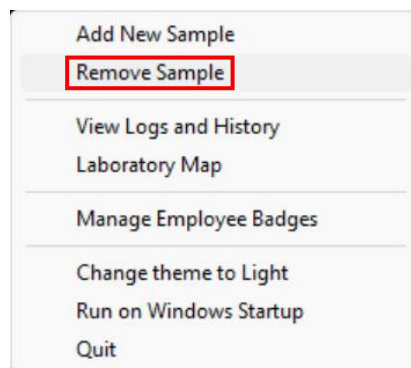
Chapter 4: Removing & Closing Samples

When testing is complete or a sample is leaving the lab, it must be removed from the active inventory. LabTrackQR includes built-in safety checks to ensure samples aren't deleted by accident.

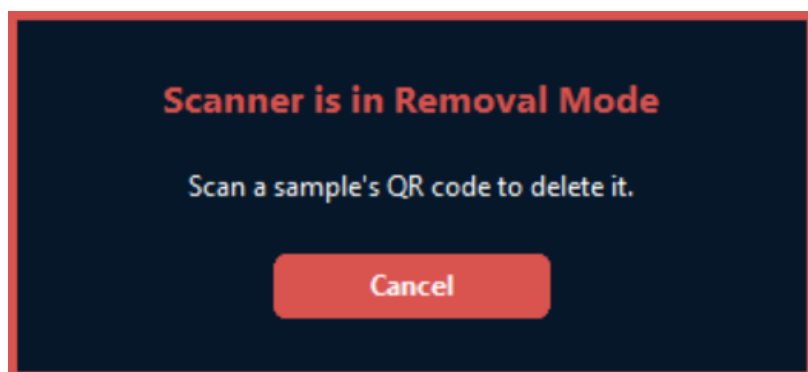
4.1 Entering Removal Mode

By default, the scanner only *adds* or *moves* samples. To delete a sample, you must explicitly tell the system you are throwing things away.

Open your System Tray menu and click **Remove Sample**.



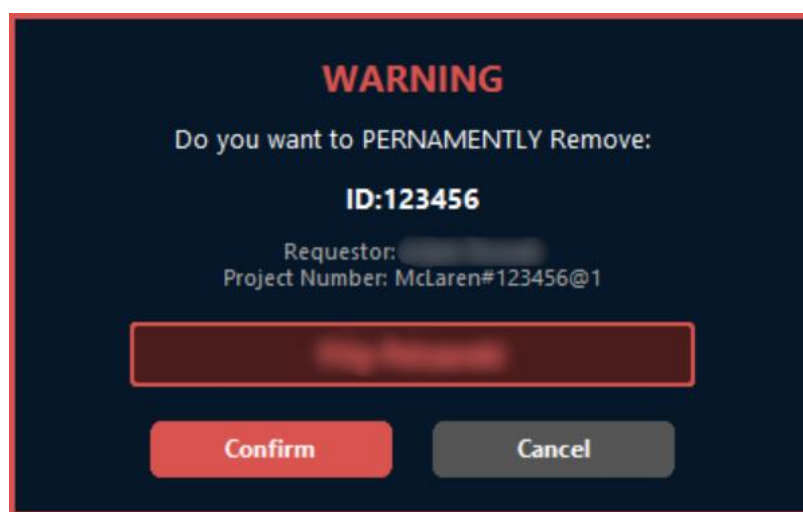
A bright red popup will appear in the center of your screen, indicating that the scanner is now armed for removal.



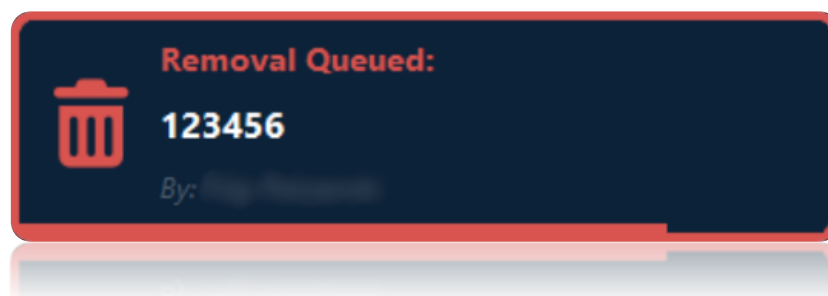
4.2 Confirming Removal

While the Removal Mode popup is on your screen, scan the barcode of the sample you want to delete.

Because deleting data is a permanent action for the active inventory, the system will immediately block the removal and display a **CRITICAL WARNING** containing the sample's Requestor and Project Number.



You must manually click the **Confirm** button using your mouse to authorize the deletion. Once confirmed, you will see a notification that the removal has been queued to the server.



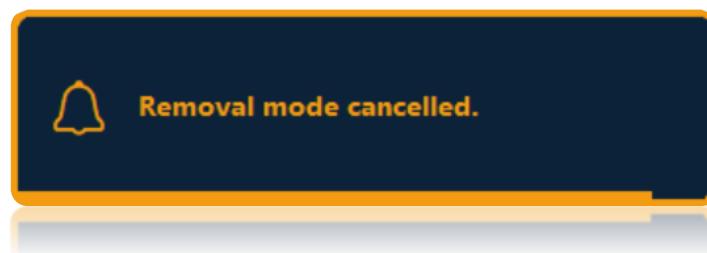
4.3 Cancelled Actions

Removal Mode is highly temporary to prevent accidents.

If you click the "Cancel" button on the warning screen, or if you simply do not click "Confirm" within 30 seconds, the system will automatically abort the deletion to protect the data.



You can also cancel out of the initial Removal Mode screen before scanning anything.



4.4 Checking Removal History

When a sample is removed, it is erased from the *Active Inventory*, but it is **not** deleted from the permanent server logs.

If you ever need to track down a deleted sample, you can look it up in the Log Viewer's history tab. The system permanently records exactly when it was removed, marking the final entry as a "Request Closed" or "Removed" action alongside the name of the user who deleted it.

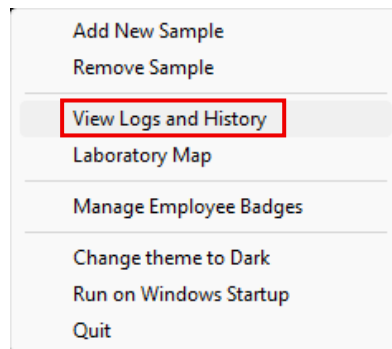
System Logs & Inventory							
View Active Inventory		Current Month Logs		Archive		Open in External Editor	
Search: mclaren		My Samples		Verification		Old Today Closed	
Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
28-09-2026	12:28:04	REQUEST CLOSED	123456		Engineering	McLaren#123456@1	

Chapter 5: The Log Viewer & Search Engine

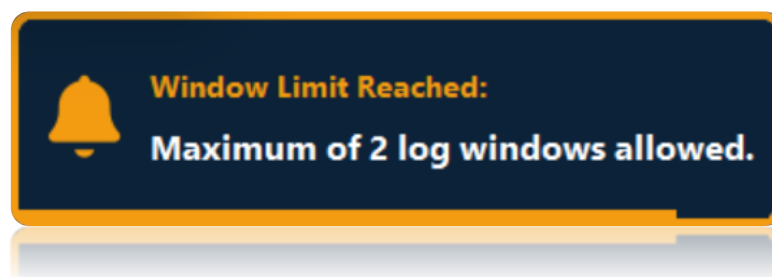
The Log Viewer is the central hub for looking up data, tracking sample history, and managing your current inventory. It is built with a powerful, Google-style search engine that can instantly filter thousands of records.

5.1 Interface & Limits

To open the viewer, right-click the System Tray icon and select **View Logs and History**.

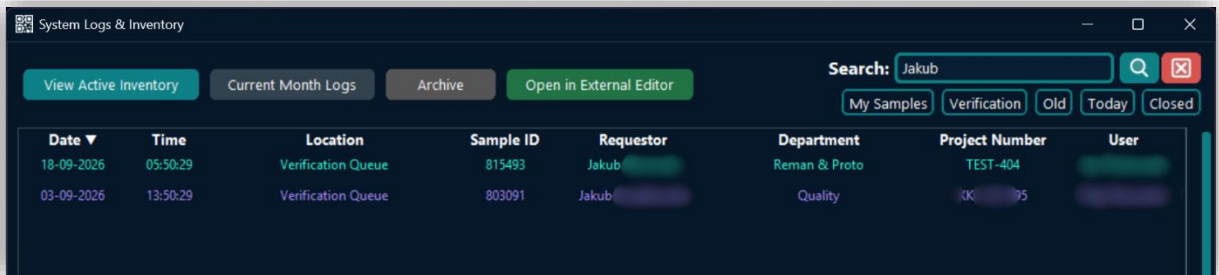


To prevent system slow-downs and keep your workspace clean, the application allows a maximum of 2 Log Viewer windows to be open at the same time. If you try to open a third, the app will alert you and simply bring your most recently used window to the front.



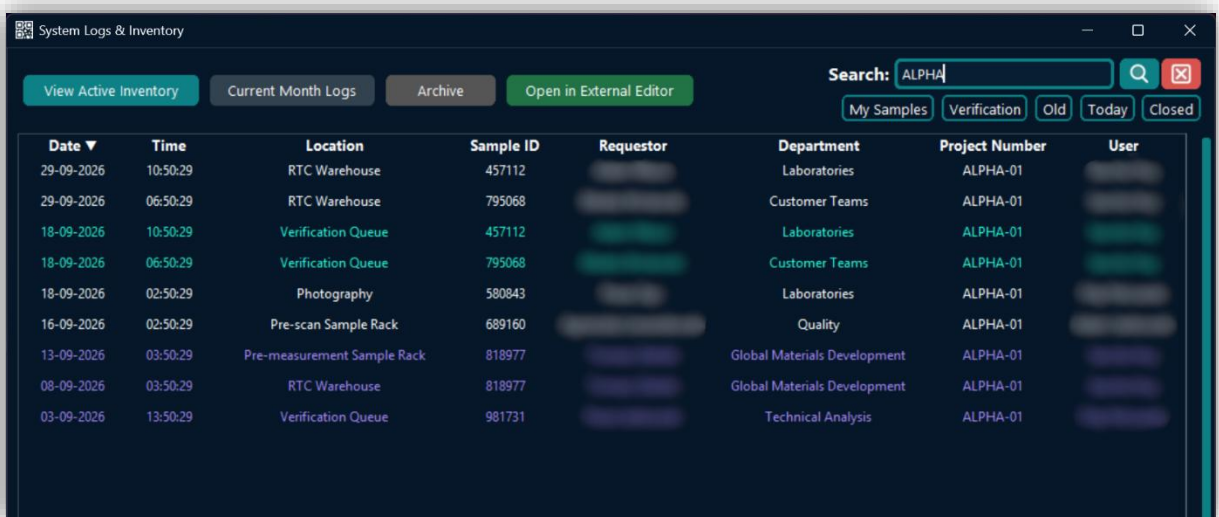
5.2 Search Functionality

The search bar at the top of the window does not just look for Sample IDs—it searches *everything*. You can type any combination of names, departments, projects, or locations to instantly narrow down the list.



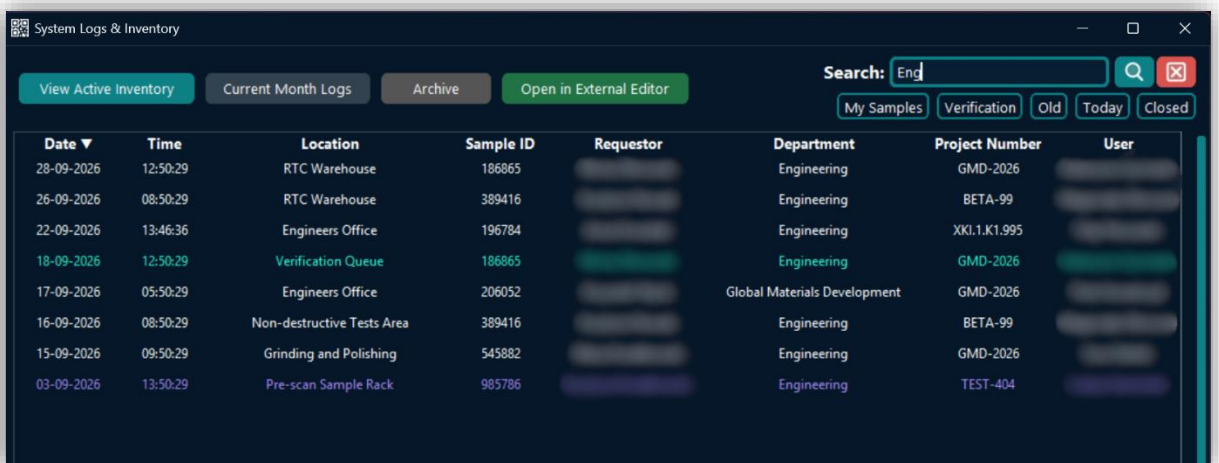
The screenshot shows the 'System Logs & Inventory' window. The search bar at the top right contains the text 'Jakub'. Below the search bar are buttons for 'My Samples', 'Verification', 'Old', 'Today', and 'Closed'. The table below displays the search results.

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
18-09-2026	05:50:29	Verification Queue	815493	Jakub	Reman & Proto	TEST-404	
03-09-2026	13:50:29	Verification Queue	803091	Jakub	Quality	(K) 95	



The screenshot shows the 'System Logs & Inventory' window. The search bar at the top right contains the text 'ALPHA'. Below the search bar are buttons for 'My Samples', 'Verification', 'Old', 'Today', and 'Closed'. The table below displays the search results.

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
29-09-2026	10:50:29	RTC Warehouse	457112		Laboratories	ALPHA-01	
29-09-2026	06:50:29	RTC Warehouse	795068		Customer Teams	ALPHA-01	
18-09-2026	10:50:29	Verification Queue	457112		Laboratories	ALPHA-01	
18-09-2026	06:50:29	Verification Queue	795068		Customer Teams	ALPHA-01	
18-09-2026	02:50:29	Photography	580843		Laboratories	ALPHA-01	
16-09-2026	02:50:29	Pre-scan Sample Rack	689160		Quality	ALPHA-01	
13-09-2026	03:50:29	Pre-measurement Sample Rack	818977		Global Materials Development	ALPHA-01	
08-09-2026	03:50:29	RTC Warehouse	818977		Global Materials Development	ALPHA-01	
03-09-2026	13:50:29	Verification Queue	981731		Technical Analysis	ALPHA-01	



The screenshot shows the 'System Logs & Inventory' window. The search bar at the top right contains the text 'Eng'. Below the search bar are buttons for 'My Samples', 'Verification', 'Old', 'Today', and 'Closed'. The table below displays the search results.

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
28-09-2026	12:50:29	RTC Warehouse	186865		Engineering	GMD-2026	
26-09-2026	08:50:29	RTC Warehouse	389416		Engineering	BETA-99	
22-09-2026	13:46:36	Engineers Office	196784		Engineering	XK1.1.K1.995	
18-09-2026	12:50:29	Verification Queue	186865		Engineering	GMD-2026	
17-09-2026	05:50:29	Engineers Office	206052		Global Materials Development	GMD-2026	
16-09-2026	08:50:29	Non-destructive Tests Area	389416		Engineering	BETA-99	
15-09-2026	09:50:29	Grinding and Polishing	545882		Engineering	GMD-2026	
03-09-2026	13:50:29	Pre-scan Sample Rack	985786		Engineering	TEST-404	

You can also combine words to create powerful shortcut searches. For example, typing a partial location and a user's name will show exactly what that specific person has in that specific area.

System Logs & Inventory

Search: micro jeol lukasz k

View Active Inventory | Current Month Logs | Archive | Open in External Editor

My Samples | Verification | Old | Today | Closed

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
17-09-2026	15:43:29	Scanning Electron Microscopy - JEOL	470515		Customer Teams	project1	lukasz K
17-09-2026	07:18:29	Scanning Electron Microscopy - JEOL	152709		Engineering		lukasz K

System Logs & Inventory

Search: global materials sample prep

View Active Inventory | Current Month Logs | Archive | Open in External Editor

My Samples | Verification | Old | Today | Closed

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
21-09-2026	15:12:18	Sample Preparation Room	666144		Global Materials Development	asdasdf	
21-09-2026	14:48:09	Sample Preparation Room	666144		Global Materials Development	asdasdf	
18-09-2026	05:34:29	Sample Preparation Room	131478		Global Materials Development	XK1.1.K1.995	
16-09-2026	15:27:29	Sample Preparation Room	762324		Global Materials Development	BETA-99	
15-09-2026	16:26:29	Sample Preparation Room	668010		Global Materials Development	project2	
15-09-2026	11:08:29	Sample Preparation Room	342963		Global Materials Development	project2	

If you want to see the entire lifecycle of a specific sample, searching its ID will display its complete timeline, from initialization in the Verification Queue all the way to its final removal.

System Logs & Inventory

Search: 666124

View Active Inventory | Current Month Logs | Archive | Open in External Editor

My Samples | Verification | Old | Today | Closed

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
21-09-2026	10:47:45	REQUEST CLOSED	666124		Laboratories	asdfasdfa	
21-09-2026	10:47:27	Engineers Office	666124		Laboratories	asdfasdfa	
21-09-2026	10:47:17	Residual Stress Measurement	666124		Laboratories	asdfasdfa	
21-09-2026	10:43:49	Scanning Electron Microscopy - JEOLSMI	666124		Laboratories	asdfasdfa	
21-09-2026	10:42:26	Metallographic Microscopy C Zeiss	666124		Laboratories	asdfasdfa	
21-09-2026	10:42:14	Metallographic Microscopy C ZeissSMP:t	666124		Laboratories	asdfasdfa	
21-09-2026	10:41:42	Stereoscopic Microscopy C Nikon	666124		Laboratories	asdfasdfa	
21-09-2026	10:38:12	Verification Queue	666124		Laboratories	asdfasdfa	

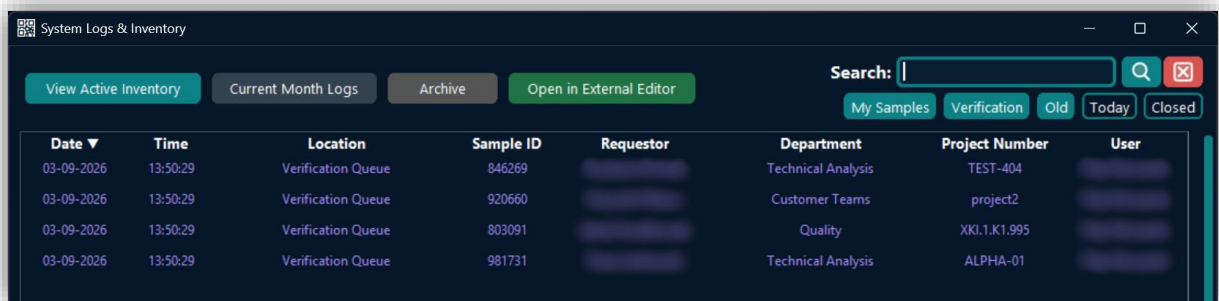
5.3 Smart Filters

Below the search bar, you will find Quick Filter buttons. These act as instant tags that you can toggle on or off. You can combine these tags with each other, or with typed keywords, to create highly specific views.



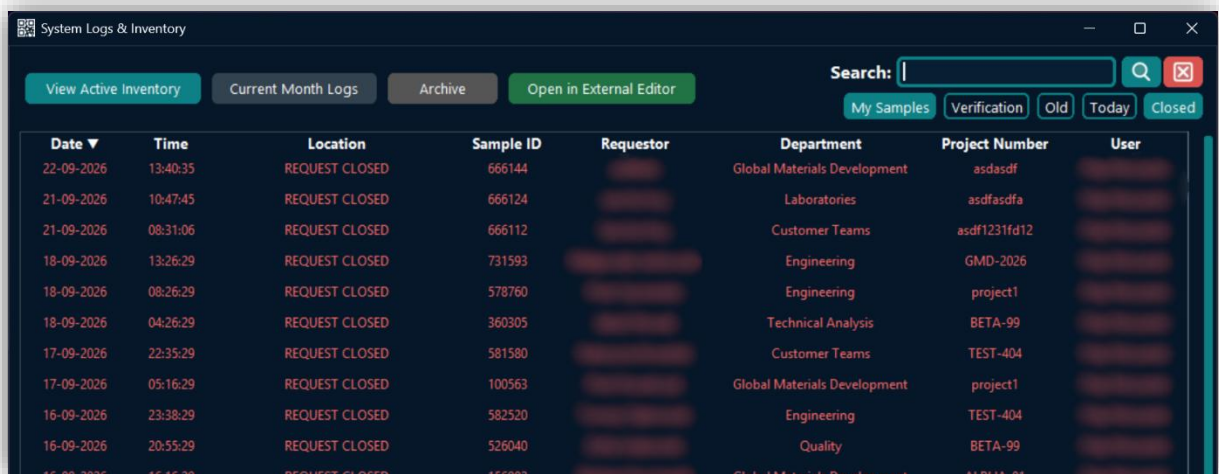
The screenshot shows the 'System Logs & Inventory' application window. The interface includes a search bar and several filter buttons: 'View Active Inventory', 'Current Month Logs', 'Archive', and 'Open in External Editor'. Below the search bar, there are additional filter buttons: 'My Samples', 'Verification', 'Old', 'Today', and 'Closed'. The table displays logs with columns: Date, Time, Location, Sample ID, Requestor, Department, Project Number, and User. The data is filtered to show logs from 18-09-2026 to 03-09-2026, all with the location 'Verification Queue'.

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
18-09-2026	12:50:29	Verification Queue	782197		Laboratories	TEST-404	
18-09-2026	12:50:29	Verification Queue	186865		Engineering	GMD-2026	
18-09-2026	10:50:29	Verification Queue	401806		Customer Teams	TEST-404	
18-09-2026	10:50:29	Verification Queue	457112		Laboratories	ALPHA-01	
18-09-2026	09:50:29	Verification Queue	860159		Global Materials Development	TEST-404	
18-09-2026	07:50:29	Verification Queue	453795		Global Materials Development	project2	
18-09-2026	06:50:29	Verification Queue	795068		Customer Teams	ALPHA-01	
18-09-2026	05:50:29	Verification Queue	815493		Reman & Proto	TEST-404	
03-09-2026	13:50:29	Verification Queue	846269		Technical Analysis	TEST-404	
03-09-2026	13:50:29	Verification Queue	920660		Customer Teams	project2	
03-09-2026	13:50:29	Verification Queue	803091		Quality	XXI.1.K1.995	
03-09-2026	13:50:29	Verification Queue	981731		Technical Analysis	ALPHA-01	



The screenshot shows the 'System Logs & Inventory' application window with the same interface as the first screenshot. The table displays logs filtered to show only those from 03-09-2026 at 13:50:29, all with the location 'Verification Queue'.

Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
03-09-2026	13:50:29	Verification Queue	846269		Technical Analysis	TEST-404	
03-09-2026	13:50:29	Verification Queue	920660		Customer Teams	project2	
03-09-2026	13:50:29	Verification Queue	803091		Quality	XXI.1.K1.995	
03-09-2026	13:50:29	Verification Queue	981731		Technical Analysis	ALPHA-01	

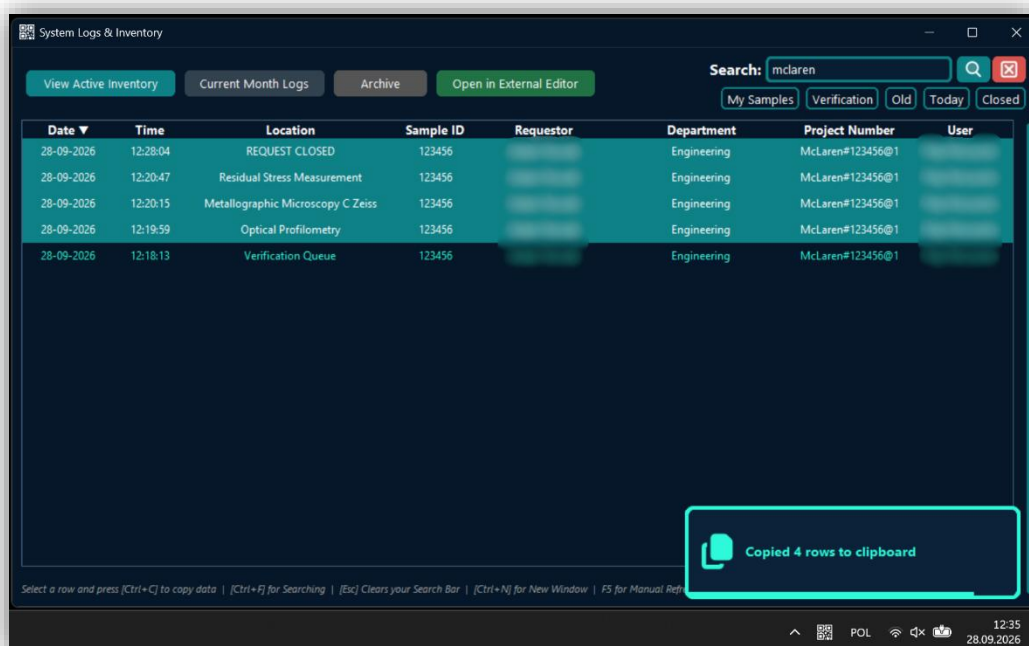
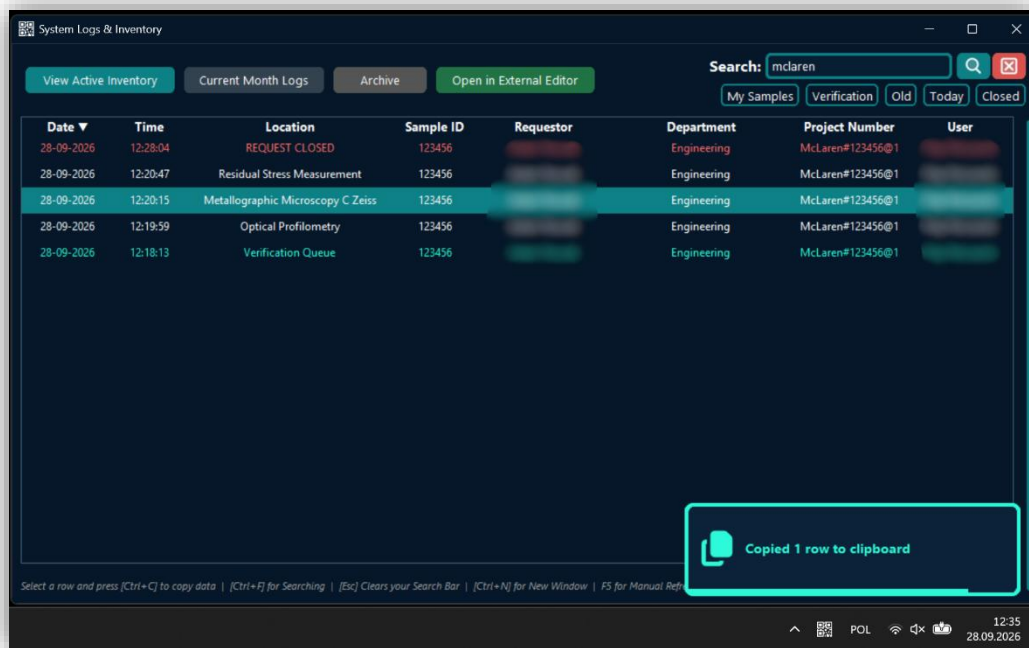


The screenshot shows the 'System Logs & Inventory' application window with the same interface as the first screenshot. The table displays logs filtered to show only those with the location 'REQUEST CLOSED'.

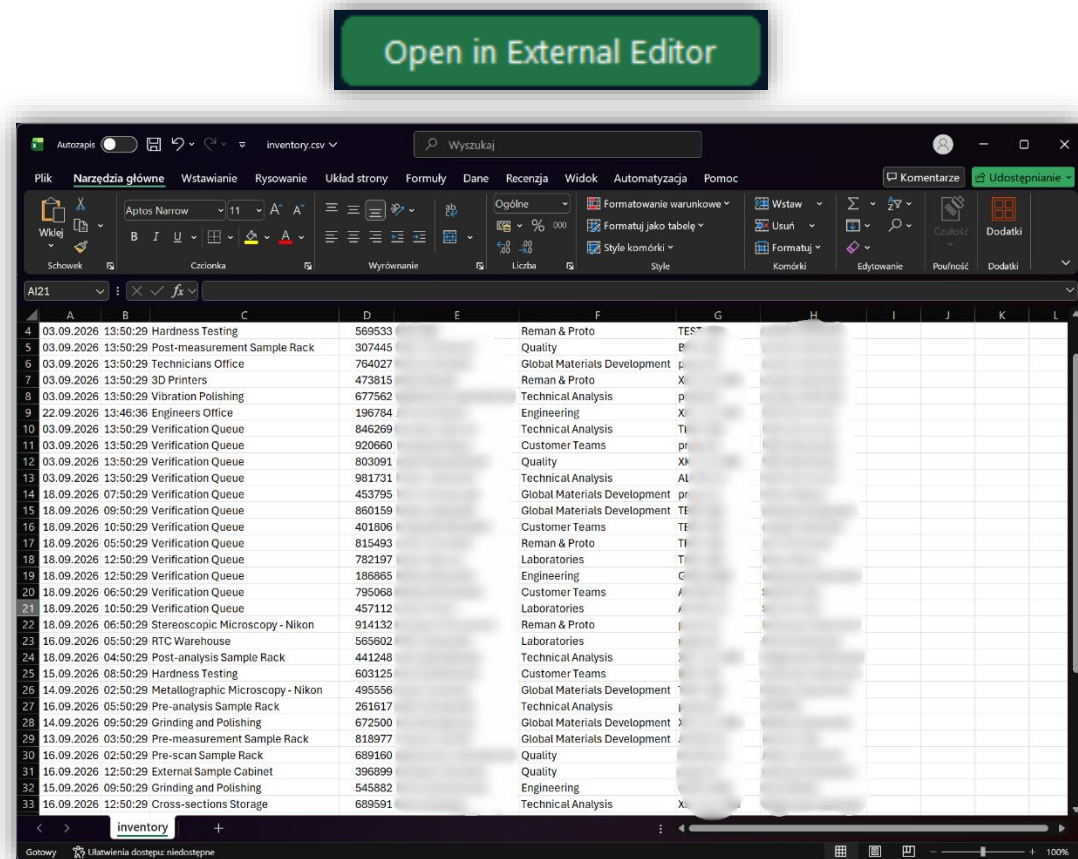
Date ▼	Time	Location	Sample ID	Requestor	Department	Project Number	User
22-09-2026	13:40:35	REQUEST CLOSED	666144		Global Materials Development	asdasdf	
21-09-2026	10:47:45	REQUEST CLOSED	666124		Laboratories	asdfasfa	
21-09-2026	08:31:06	REQUEST CLOSED	666112		Customer Teams	asdf1231fd12	
18-09-2026	13:26:29	REQUEST CLOSED	731593		Engineering	GMD-2026	
18-09-2026	08:26:29	REQUEST CLOSED	578760		Engineering	project1	
18-09-2026	04:26:29	REQUEST CLOSED	360305		Technical Analysis	BETA-99	
17-09-2026	22:35:29	REQUEST CLOSED	581580		Customer Teams	TEST-404	
17-09-2026	05:16:29	REQUEST CLOSED	100563		Global Materials Development	project1	
16-09-2026	23:38:29	REQUEST CLOSED	582520		Engineering	TEST-404	
16-09-2026	20:55:29	REQUEST CLOSED	526040		Quality	BETA-99	
16-09-2026	16:16:30	REQUEST CLOSED	155093		Global Materials Development	ALPHA-01	

5.4 Exporting & Copying

If you need to extract data for reports or emails, you don't need to open external files. Simply select the rows you want (holding Shift or Ctrl allows multi-selection) and press **Ctrl+C** on your keyboard. A notification will confirm the rows have been copied to your clipboard, perfectly formatted to paste into Excel or Teams.

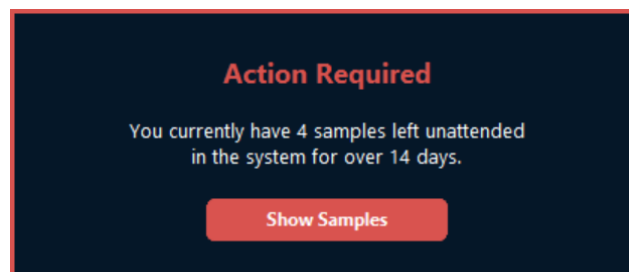


If you do need to view the raw master files, click the **Open in External Editor** button at the top to open the current data directly in Excel.



5.5 The 14-Day Warning System

To ensure the laboratory doesn't get cluttered with forgotten samples, the system actively monitors the age of all items. If you have samples sitting untouched in the active inventory for over 14 days, the app will hit you with an "Action Required" hard-stop popup when you log in.

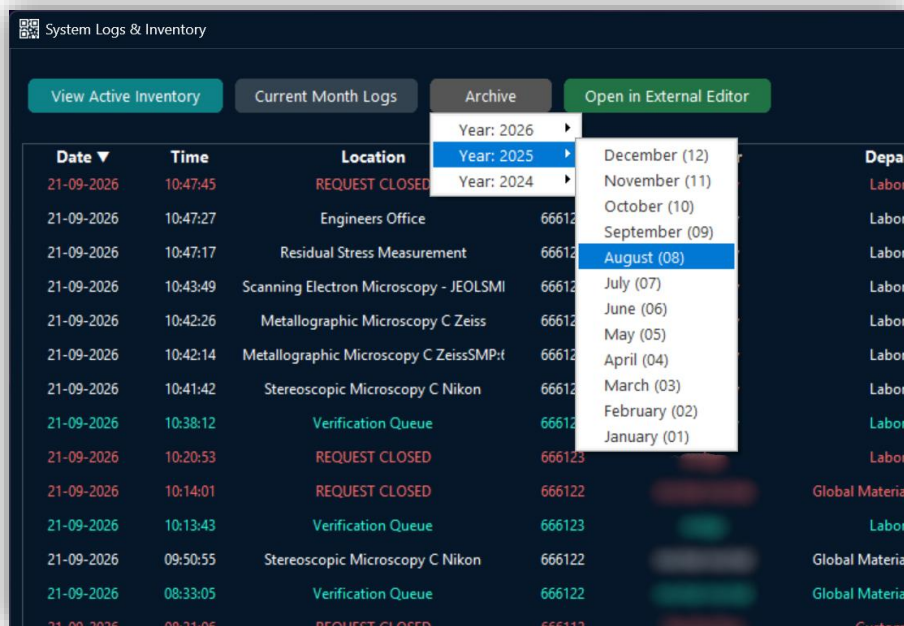


Clicking "Show Samples" on this popup will automatically open the Log Viewer with the "My Samples" and "Old" filters already activated, showing you exactly what needs your attention.

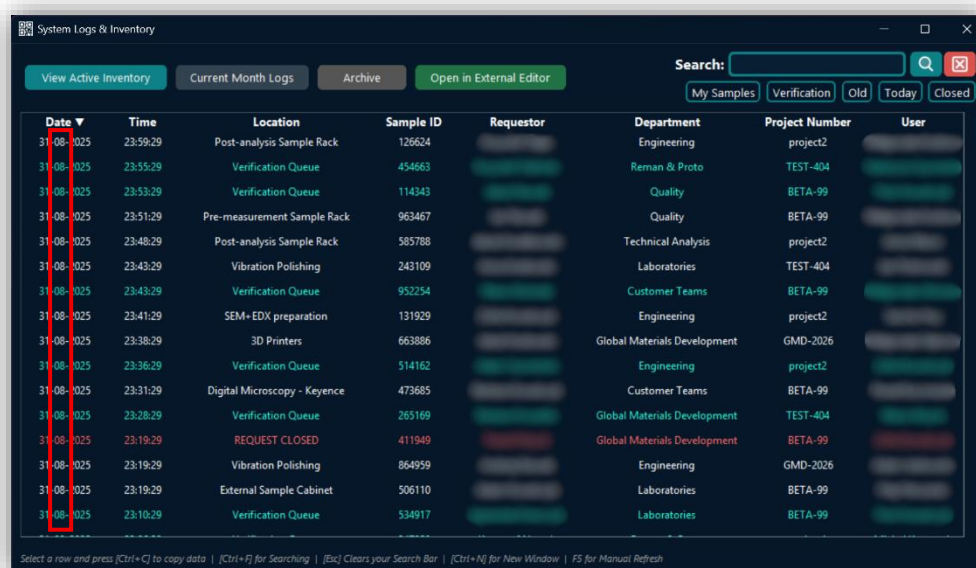
5.6 Using the Archive

The Log Viewer defaults to the active inventory, but it can pull historical data seamlessly.

Click the **Archive** button to reveal a dropdown menu organized by year and month.



Selecting a past month will pull those exact logs from the server so you can search and filter them just like live data.

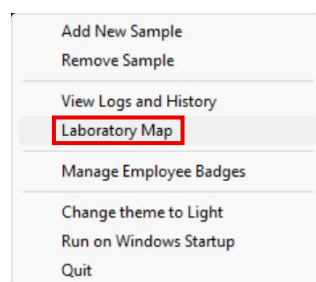


Chapter 6: The Laboratory Map Viewer

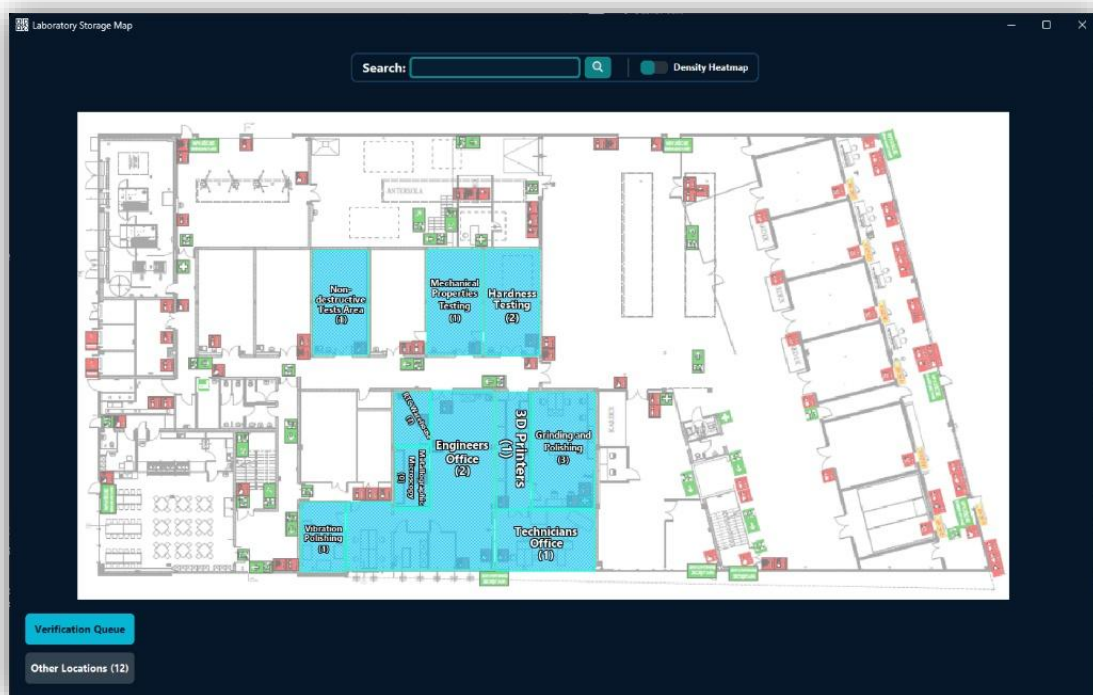
The Map Viewer provides a visual, interactive layout of the laboratory, allowing you to see exactly where samples are physically located in real-time.

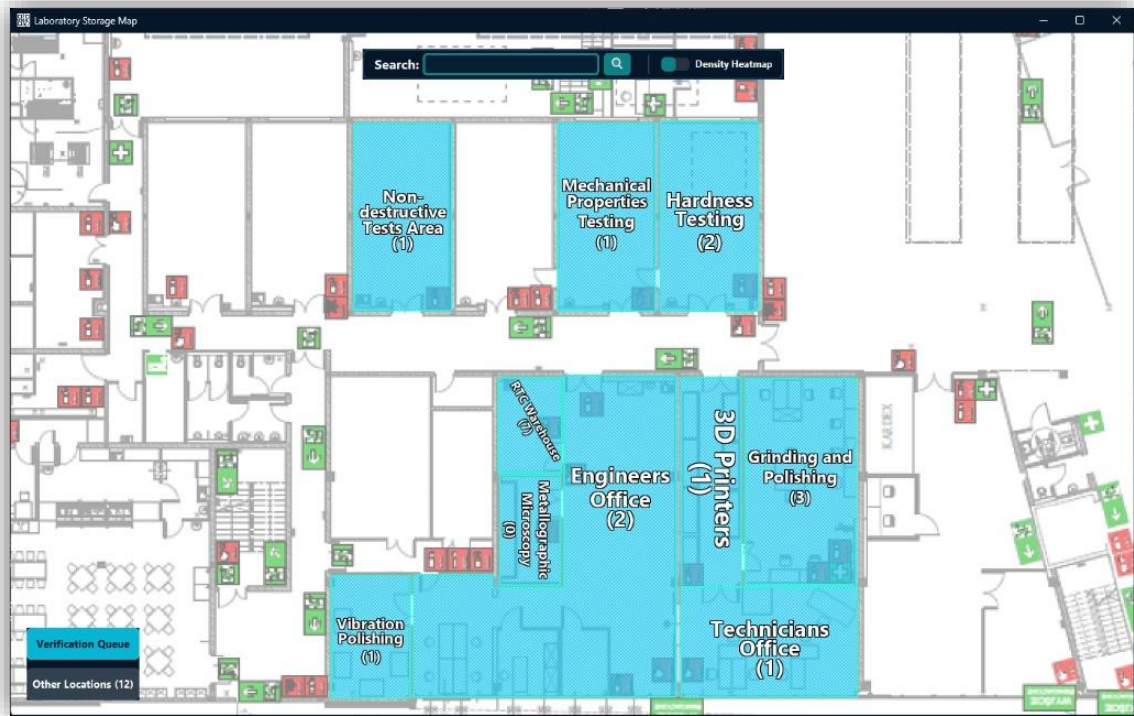
6.1 Map Navigation

Open the System Tray menu and select **Laboratory Map**.



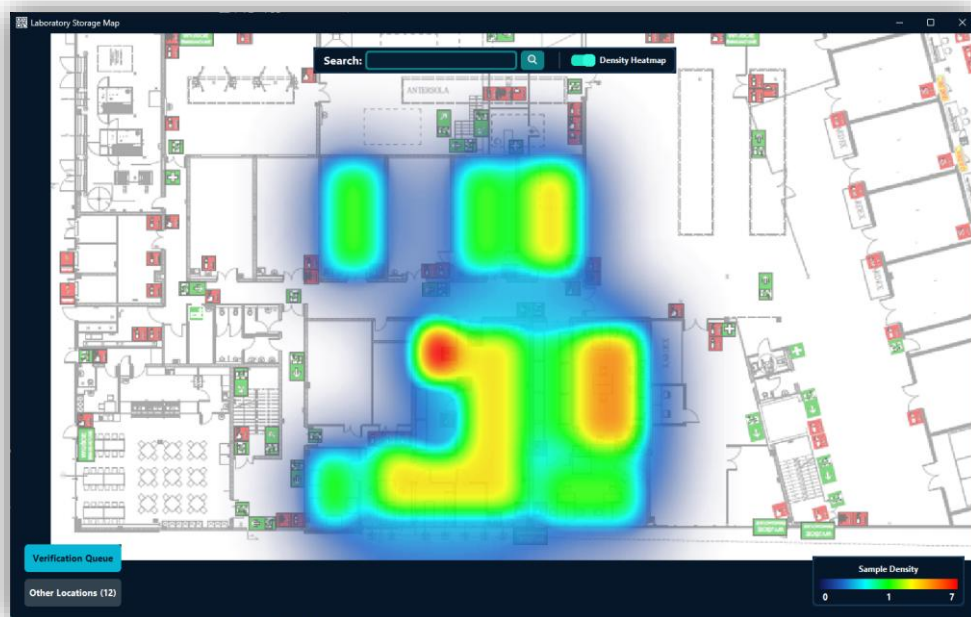
The interface allows you to pan around the facility by clicking and dragging with your mouse (Middle-Click or Left-Click). You can zoom in and out using **Ctrl + Mouse Wheel**. A helpful minimap in the bottom corner shows your current viewport in relation to the whole building.





6.2 Density Heatmap

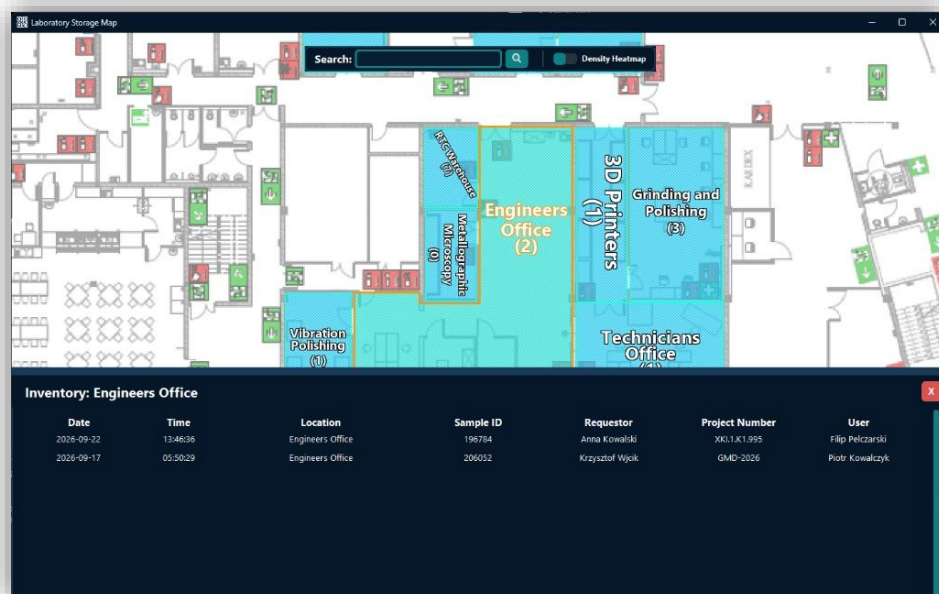
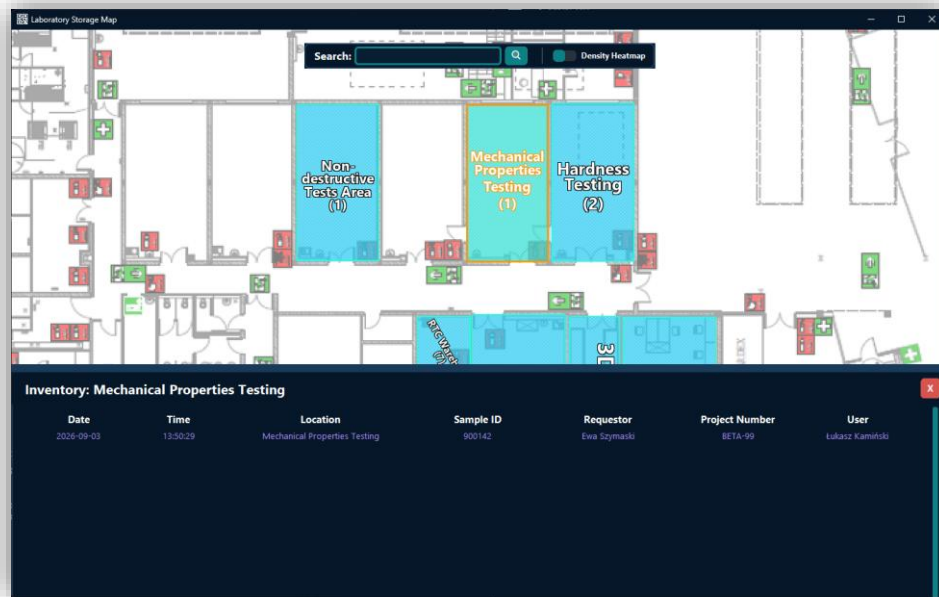
To instantly see which areas of the lab are currently holding the most items, toggle the **Density Heatmap** switch at the top. The zones will change color based on their inventory load, shifting from green (low density) to red (high density).



6.3 Locating Samples

You can interact with the map to view inventory in two ways:

1. **Clicking on Zones** Clicking any highlighted zone on the map will slide up the Bottom Sheet, displaying a live table of every sample currently sitting in that physical location.



2. **Radar Search** If you have a Sample ID and need to find it, type it into the Search bar at the top of the map.



If the sample is located in a mapped zone, the map will instantly jump to that area and pulse the zone to show you where it is. If the sample is waiting to be processed, or sitting in an area that isn't drawn on the map, the Bottom Sheet will slide up automatically to show you, its status.

Samples in Unmapped Locations

Date	Time	Location	Sample ID	Requestor	Project Number	User
2026-09-03	13:50:29	Pre-scan Sample Rack	985786	Krystyna Kwiatkowski	TEST-404	Łukasz Kamiński
2026-09-16	02:50:29	Pre-scan Sample Rack	689160	Agnieszka Lewandowski	ALPHA-01	Adam Jankowski
2026-09-03	13:50:29	Post-measurement Sample Rack	307443	Piotr Grabowski	BETA-99	Łukasz Kamiński
2026-09-18	06:50:29	Stereoscopic Microscopy - Nikon	914132	Krystyna Kaczmarek	project2	Katarzyna Szymanski
2026-09-18	04:50:29	Post-analysis Sample Rack	441248	Jan Lewandowski	XXL1.X1.995	Magorzata Dbrowski
2026-09-14	02:50:29	Metallographic Microscopy - Nikon	495556	Adam Kowalski		
2026-09-16	05:50:29	Pre-analysis Sample Rack	261617	Zofia Grabowski		
2026-09-13	03:50:29	Pre-measurement Sample Rack	818977	Tomasz Zielski		
2026-09-16	12:50:29	External Sample Cabinet	396899	Krystyna Kozowski		

Found in Unmapped Locations:
12333

Chapter 7: Server Administration & Setup

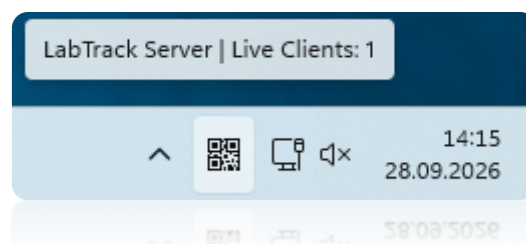
The LabTrackQR Server is the backbone of the system. It handles all data synchronization between scanning stations, actively manages employee credentials, and runs automated, self-healing backups to protect the laboratory's inventory.

7.1 Server Basics

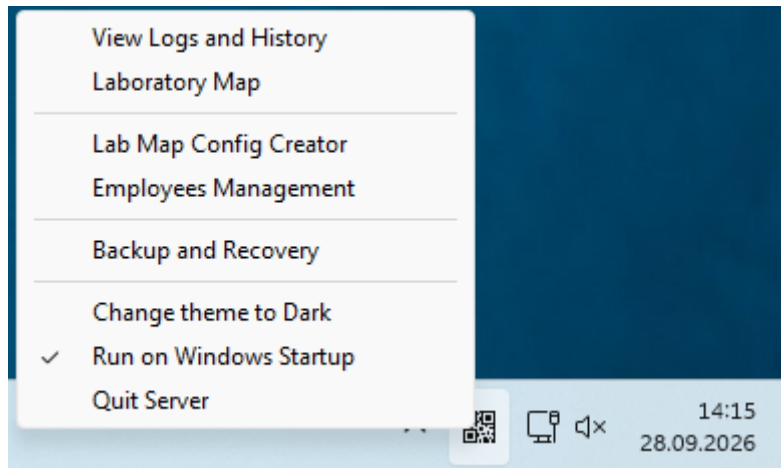
When you launch the Server application, it will display a brief startup screen while connecting to the master database.



Just like the client app, the server runs quietly in the Windows System Tray. By hovering your mouse over the server icon, you can instantly see a live counter of how many client scanning stations are currently connected and active.



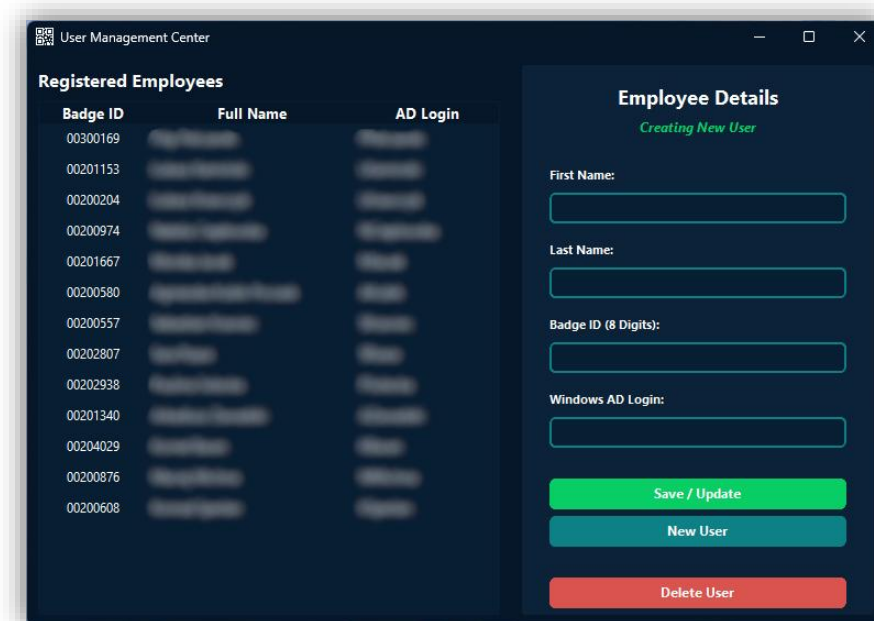
Right-clicking the icon opens the Server control panel, giving administrators direct access to management tools and global settings.



7.2 Server User Management

Adding or removing employees from the system must be done through the Server.

Open the Server Tray Menu and click **Employees Management**.



This opens the User Management Center. On the left, you will see a complete, sortable database of all registered employees and their assigned badge IDs. On the right is the Employee Details form.

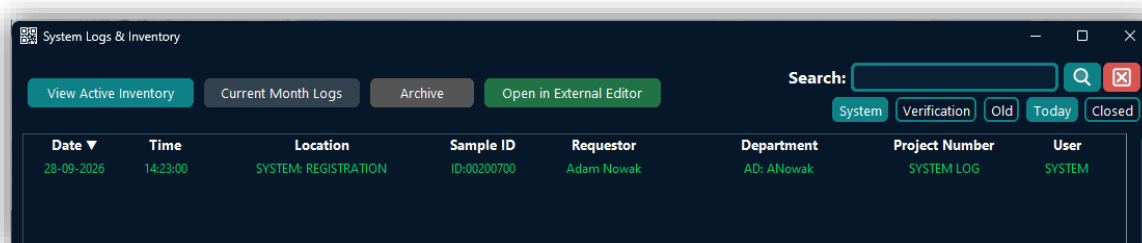
The screenshot shows the 'User Management Center' window. On the left, the 'Registered Employees' table has three columns: 'Badge ID', 'Full Name', and 'AD Login'. It contains a list of employees. On the right, the 'Employee Details' form is titled 'Editing Existing User'. It includes input fields for 'First Name', 'Last Name', 'Badge ID (8 Digits)', and 'Windows AD Login'. Below these fields are three buttons: 'Save / Update' (green), 'New User' (teal), and 'Delete User' (red).

To add someone new, fill out their First Name, Last Name, an 8-digit Badge ID, and their Windows AD Login. Once you click "Save / Update", they are instantly added to the active database on the left.

This screenshot shows the 'User Management Center' window with the 'Employee Details' form titled 'Creating New User'. The 'First Name' field contains 'Adam', 'Last Name' contains 'Nowak', 'Badge ID (8 Digits)' contains '00200700', and 'Windows AD Login' contains 'ANowak'. The 'Save / Update' button is green, 'New User' is teal, and 'Delete User' is red. The 'Registered Employees' table on the left is also visible.



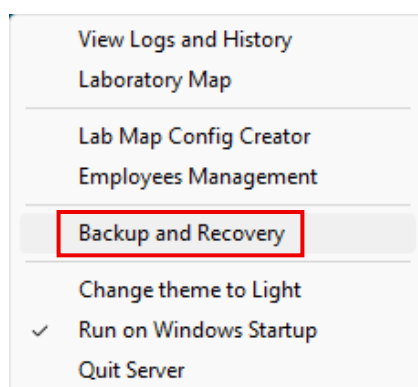
Note: For security and auditing, the Server automatically logs the creation and deletion of users. You can view these admin actions in the Log Viewer by filtering for the "System" tag.



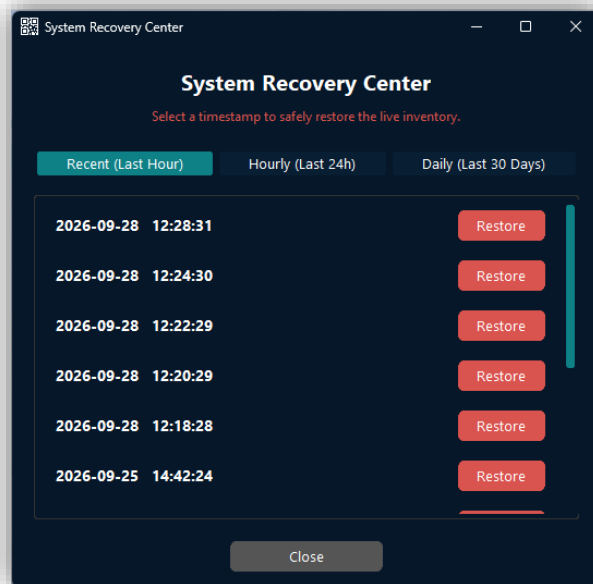
7.3 Backup & Recovery Center

The Server automatically zips and backs up the entire database every few minutes. If a file is ever corrupted, accidentally deleted, or aggressively modified, you can roll back time.

Open the Server Tray Menu and select **Backup and Recovery**.



The System Recovery Center organizes automated backups into three separate tiers to ensure you always have a safe restore point without cluttering the hard drive:

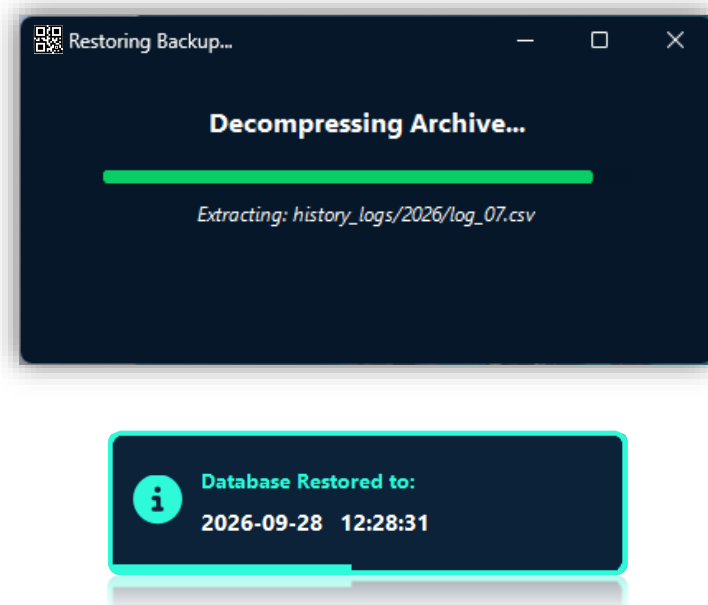


- Recent (Last Hour): Snapshots taken every few minutes.
- Hourly (Last 24h): Pinned snapshots at the top of every hour.
- Daily (Last 30 Days): Permanent daily records.

When you click "Restore" on any backup, the Server will halt and issue a **CRITICAL WARNING**, as this will completely overwrite the live inventory.



Once you confirm, the system will decompress the archive and rebuild the files. A final popup will confirm the successful database restoration.



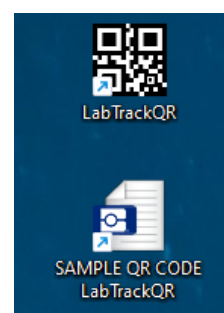
Chapter 8: Hardware Prep (Brady Workstation)

For the system to recognize samples, physical items must be tagged with correctly formatted QR codes. This is managed using the Brady Workstation software and a Brady M710 printer.

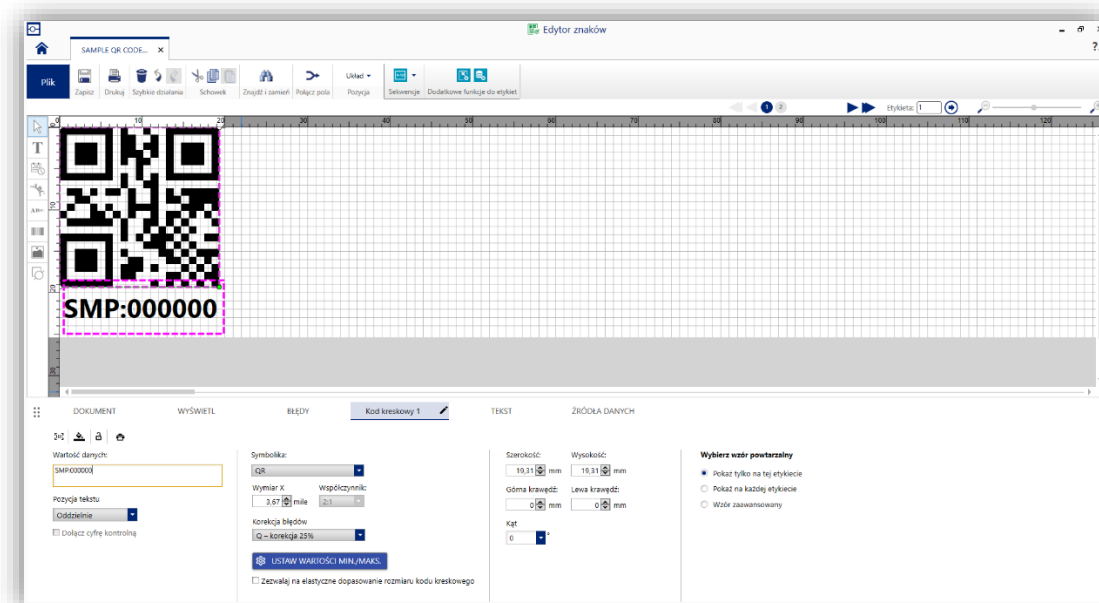
8.1 Setting Up the Template

We use a standardized Brady Workstation template to ensure the QR codes are generated correctly for the scanners.

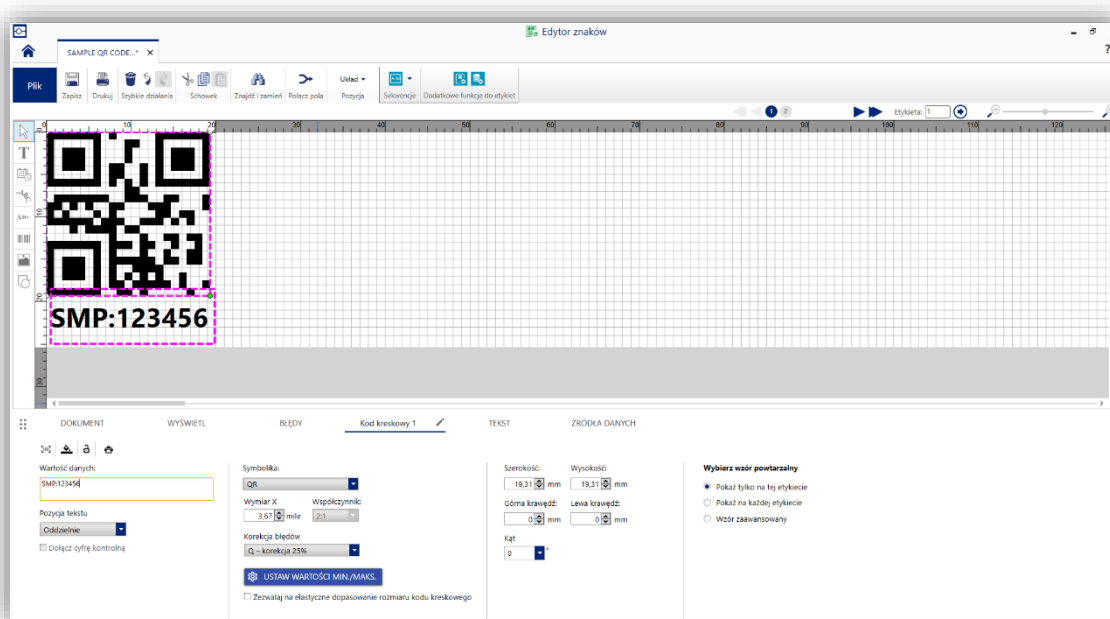
Open the pre-configured template directly from your desktop.



The QR code value must *always* begin with the prefix **SMP:** followed by the unique ID number. For example, if you open a fresh template, it might default to SMP:000000.



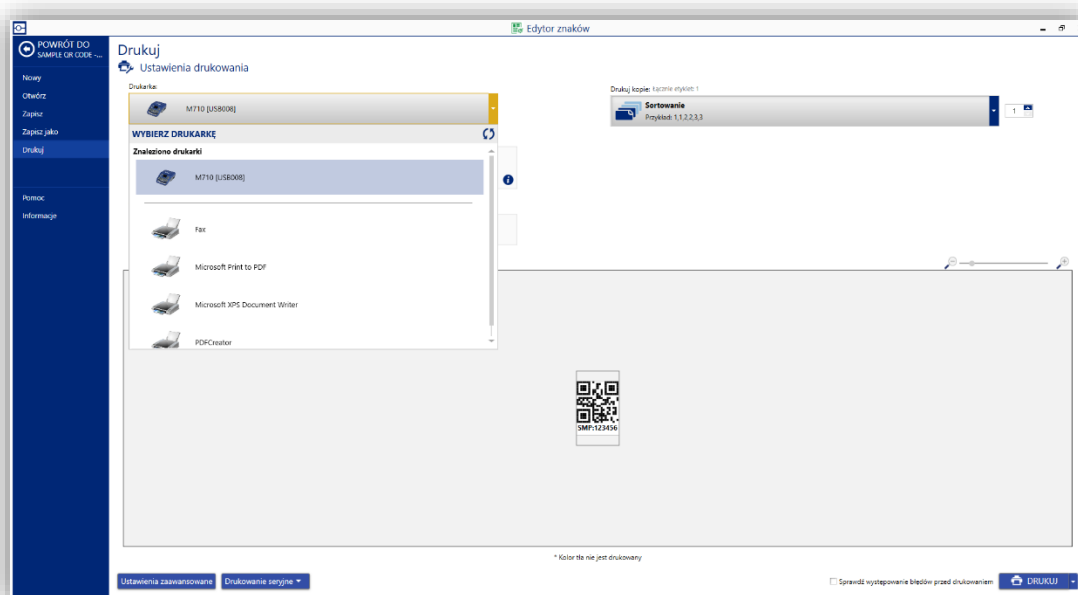
Simply click the QR code to edit its value and input the new 6-digit sample ID (e.g., SMP:123456).



8.2 Printing Labels

Once the data is entered and verified on the screen, proceed to print.

Ensure the correct printer (e.g., Brady M710) is selected from the printer dropdown menu in the upper left corner.



Verify your print settings, quantity, and layout, then click print to generate the physical QR tags.

